

Code No: 114CQ**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech II Year II Semester Examinations, September/ October - 2023****DATABASE MANAGEMENT SYSTEMS****(Computer Science and Engineering)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A**(25 Marks)**

- 1.a) Define DBMS. [2]
- b) Write the purpose of DDL. List any 2 commands of DDL. [3]
- c) What is tuple relational calculus? [2]
- d) Write the purpose of order by in SQL with example. [3]
- e) What is functional dependency? [2]
- f) What is BCNF? [3]
- g) Define transaction in DBMS. [2]
- h) Explain serializability in databases [3]
- i) What is secondary indexes? [2]
- j) Write about tree structured indexing. [3]

PART - B**(50 Marks)**

- 2.a) List the applications of DBMS.
 - b) Explain various data models. [5+5]
- OR**
- 3.a) Write about Relational model in DBMS.
 - b) Write a note on views. [5+5]
- 4.a) Explain about the set comparison operators used in SQL queries with example.
 - b) How to perform comparison using NULL values? Explain. [5+5]
- OR**
5. Elaborate on joins in SQL queries with example. [10]
 6. Discuss Multi-valued dependency and Fourth Normal form with suitable example. [10]
- OR**
- 7.a) Write detailed notes on the dependency preserving decomposition.
 - b) Explain fifth Normal form. [5+5]
- 8.a) What is concurrency control in databases? Explain Time stamp-based protocol in detail.
 - b) Explain multiple granularity. [5+5]
- OR**
9. Why we need to backup database? Discuss the remote backup systems of the DBMS. [10]
 10. Elaborate on file organization of the DBMS. [10]
- OR**
11. Discuss on any two methods of hash based indexing. [10]